

★ WEEK 9 · PART 3 · PRACTICE

One Filter, Two Uses - Worked by Hand

An Adaptive Pairs Signal, and a Nowcast Through Missing Data

The Kalman filter, applied to trading and to the economy

Part 3's applications, with the arithmetic shown. We turn a drifting hedge ratio into a live trading signal, then use a dynamic factor model to nowcast a hidden state of the economy - even when an indicator is missing.

3P

PART 3 · APPLICATIONS

Two Uses, Computed by Hand

Case A: an adaptive spread from a drifting hedge ratio, turned into a z-score and a signal - and why a fixed hedge fails. Case B: a nowcast from a dynamic factor model, filling the ragged edge of missing data. Then a Claude Code lab.

What Part 3 Told Us

Now we apply the filter - with numbers

1

The shared capability is denoising

Both uses pull a clean hidden state from noisy data - a fair hedge ratio, or the true state of the economy.

2

Pairs trading: an adaptive hedge

Let the hedge ratio drift as a state, and the spread stays mean-reverting. We turn it into a live signal by hand.

3

Nowcasting: a hidden factor

One factor drives many indicators arriving at different times. The filter reads the present, even with data missing.

Everything here is a calculator exercise: one spread and z-score, one factor from two indicators.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the filtered hidden state.
- Compute the signal or the nowcast by hand.
- Then read the decision it supports.

What you should watch

- How an adaptive spread beats a fixed one.
- How a z-score becomes a trade.
- How the filter handles a missing indicator.

An Adaptive Pairs Signal

A drifting hedge, a live trade

In Week 8 we filtered a hedge ratio; now we trade on it. As the hedge ratio drifts, the filter keeps it current, so the spread stays mean-reverting. We compute today's adaptive spread, turn it into a z-score and a signal, and show that a fixed hedge ratio would have drifted off completely

The inputs

- Today's filtered hedge ratio beta = 0.90.
- Prices $A = 52.0$, $B = 57.0$.
- Recent spread mean 0.20, sd 0.25.

What we'll do by hand

- Form the adaptive spread and its z-score.
- Read off the trading signal.
- Compare against a stale fixed hedge.

The Adaptive Spread and Signal

A drifting hedge keeps the spread honest

$$\text{adaptive spread} = A - \beta_t B, \quad z = \frac{\text{spread} - \mu}{\sigma} \quad (\beta_t \text{ drifts, kept live by the filter})$$

The spread

- A minus the live hedge ratio times B.
- Because beta is current, the spread reverts.
- A fixed beta would let it wander.

The signal

- Standardize the spread into a z-score.
- Beyond +/- 2, the spread is stretched.
- Trade against the stretch, expect reversion.

Today's Spread and Signal

From a live hedge ratio to a trade

$$\text{spread} = 52.0 - 0.90(57.0) = 0.70$$

$$z = \frac{0.70 - 0.20}{0.25} = \mathbf{2.0} \quad \Rightarrow \quad |z| \geq 2 : \text{SHORT the spread}$$

The adaptive spread is 0.70, a z-score of exactly 2.0 - stretched two standard deviations rich, so the signal is to short the spread and wait for reversion.

Why a Fixed Hedge Fails

The same prices, a stale ratio

$$\text{fixed } \beta = 0.80 : \text{spread} = 52.0 - 0.80(57.0) = 6.40 \Rightarrow \text{miscalibrated, not tradable}$$

The fixed hedge

- An old beta of 0.80 gives a spread of 6.40.
- That's nowhere near the mean of 0.20.
- The z-score would be enormous - and useless.

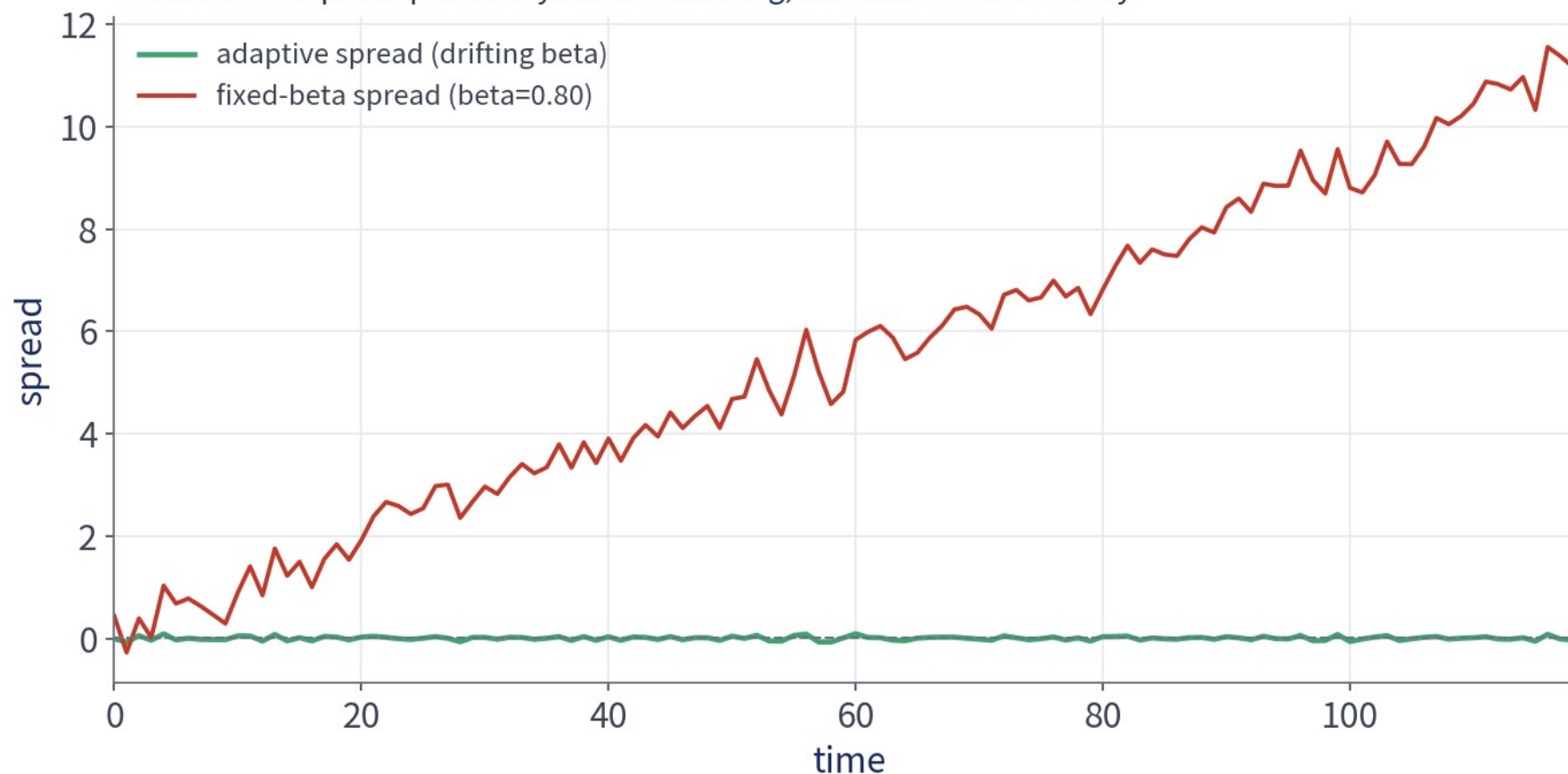
The lesson

- The relationship drifted; the fixed hedge didn't.
- Its spread is miscalibrated, not tradable.
- The live, filtered hedge keeps the trade honest.

Adaptive Stays Honest; Fixed Drifts

The adaptive spread (green) hugs its mean and oscillates - exactly what a pairs trade needs. The fixed-beta spread (red) drifts away as the hedge ratio changes, so its z-scores become meaningless. A live hedge keeps the signal tradable

Case A: the adaptive spread stays mean-reverting; the fixed one drifts away



The Filter Keeps the Trade Alive

Denoising, applied to a spread

Without the filter

- A one-time regression sets the hedge.
- It goes stale as the market shifts.
- The spread drifts, the signal breaks.

With the filter

- The hedge ratio updates every day.
- The spread stays centered and mean-reverting.
- The z-score remains a valid signal.

Nowcasting a Hidden Factor

Reading the present through missing data

The same filter reads the economy. A single hidden factor - the state of activity - drives many indicators that arrive at different times, so today some are missing (the ragged edge). The filter combines a prior with whatever indicators are available, dropping the missing one to produce a nowcast. We compute it by hand

The setup

- A hidden factor with persistence $\phi = 0.8$.
- Three indicators load on it (1.0, 1.2, 0.8).
- This period, indicator 3 is missing.

What we'll do by hand

- Predict the factor forward.
- Turn each available indicator into a read.
- Combine them into one nowcast.

The Dynamic Factor Model

One factor, many indicators

$$\begin{array}{ll} \text{state: } f_t = \phi f_{t-1} + \text{noise} & \text{(hidden factor)} \\ \text{observation: } y_{i,t} = \lambda_i f_t + \varepsilon_{i,t} & \text{(each indicator loads on } f) \end{array}$$

The state

- A hidden factor that persists over time.
- It's the common driver behind the data.
- The thing we want to nowcast.

The observations

- Each indicator is the factor times a loading.
- Plus its own measurement noise.
- Some arrive late or not at all.

Each Indicator Implies a Factor

Predict, then read the available data

$$\begin{aligned} \text{predict: } f^- &= \phi f_{t-1} = 0.8(0.30) = 0.24 \\ \text{indicator 1: } \hat{f} &= \frac{0.5}{1.0} = 0.50, \quad \text{indicator 2: } \hat{f} = \frac{0.7}{1.2} = 0.58, \quad \text{indicator 3: missing} \end{aligned}$$

Predict the factor forward to 0.24. Indicators 1 and 2 imply factor values of 0.50 and 0.58; indicator 3 is missing, so it simply drops out of this update.

Combine Into a Nowcast

Precision-weighted, missing dropped

$$\text{nowcast } f = \frac{\text{prior} \cdot p_0 + \hat{f}_1 p_1 + \hat{f}_2 p_2}{p_0 + p_1 + p_2} = \mathbf{0.47} \quad (\text{missing indicator simply drops out})$$

The combination

- Weight the prior and each read by precision.
- More precise reads count for more.
- The nowcast lands at 0.47.

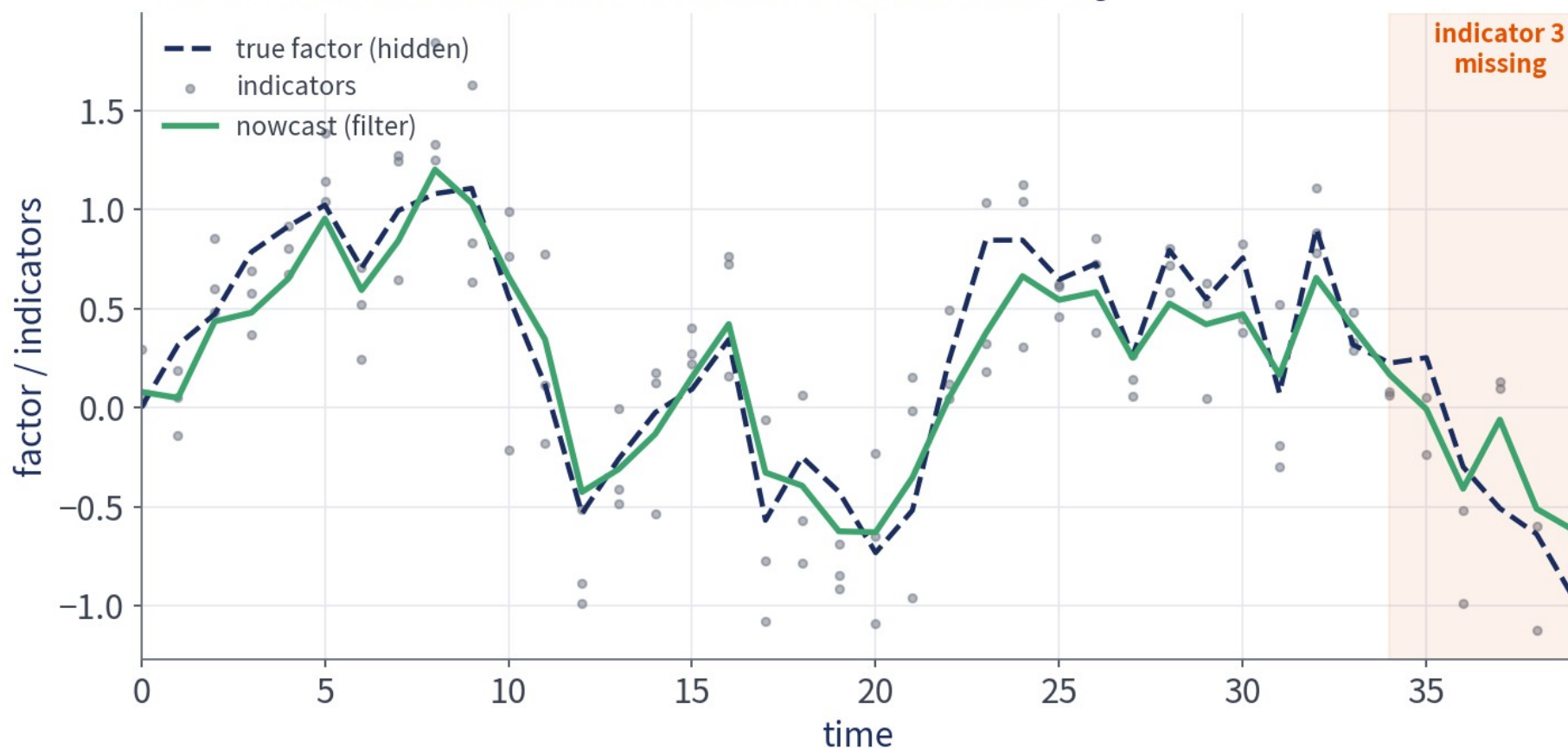
The ragged edge

- Indicator 3 contributes nothing this period.
- The filter uses only what's available.
- No imputation, no dropped update.

The Nowcast Tracks Through the Gap

The nowcast (green) follows the hidden factor (dashed) by pooling the noisy indicators. In the shaded stretch, indicator 3 is missing - yet the nowcast keeps tracking, because the filter simply weights whatever data has arrived. That is the ragged edge handled.

Case B: the nowcast tracks the factor - even where an indicator is missing



Why the Filter Fits Nowcasting

Missing data is not a problem

The hard part of nowcasting

- Indicators arrive weekly, monthly, quarterly.
- At any moment, some are not yet in.
- You still need a read on the present.

The filter's answer

- Update from whatever has arrived.
- Missing rows just drop out of the step.
- The nowcast is always current.

One Filter, Two Uses

The same engine, two problems

A pairs trade and a nowcast look nothing alike, yet both are the Kalman filter denoising a hidden state. In one, the state is a hedge ratio and the output is a trade; in the other, the state is the economy and the output is a real-time read. Same predict-update engine, different state

Pairs trading

- State: the drifting hedge ratio.
- Output: an adaptive spread and z-score.
- Value: a signal that stays valid.

Nowcasting

- State: the hidden factor of activity.
- Output: a real-time estimate of the present.
- Value: robustness to missing data.

Reproduce Both Applications

Direct Claude Code to check your arithmetic

GOAL Verify the adaptive signal and the nowcast, then run both on real data.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - adaptive signal:

- * from $\beta=0.90$, $A=52.0$, $B=57.0$, print the spread, the z-score (mean 0.20, sd 0.25), and the signal;
- * compare to a fixed $\beta=0.80$ spread.

CASE B - nowcast:

- * predict the factor ($\phi=0.8$, prior 0.30);
- * from indicators 0.5, 0.7 (loadings 1.0, 1.2) with indicator 3 missing, print the nowcast.

Then run both on real series and plot the results.

Match every printed number to the slides.

Check against the slides

- adaptive spread 0.70, $z = 2.0$.
- fixed-beta spread 6.40.
- factor predict 0.24.
- nowcast 0.47.

Deliverable: `part3_practice.py` whose printed numbers match the signal and the nowcast.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why does an adaptive spread beat a fixed one?

The hedge drifts; the filter keeps it current

Q2 How does the filter handle a missing indicator?

It drops out; the update uses what's available

Q3 What do both applications have in common?

Denoising a hidden state, same filter

All three come from the adaptive-spread signal and the ragged-edge nowcast.

Three Pitfalls to Avoid

Where the applications slip

Common mistakes

- "A fixed hedge ratio is good enough."
- "Impute or drop the whole period if data is missing."
- "The nowcast needs every indicator."

The reality

- A drifting relationship needs a live hedge.
- The filter uses whatever has arrived - no imputation.
- A partial read still gives a valid nowcast.

Two Applications to Remember

Part 3 applications in one screen

1

An adaptive spread makes a live signal

With a filtered hedge ratio of 0.90, the spread was 0.70 and the z-score exactly 2.0 - a short signal. A stale fixed beta of 0.80 gave a spread of 6.40, miscalibrated and untradable. The filter keeps the trade honest.

2

A nowcast reads the present through gaps

A dynamic factor model predicted the factor to 0.24, then combined two available indicators (0.50 and 0.58) by precision into a nowcast of 0.47. The missing indicator simply dropped out - the ragged edge, handled.

3

One filter, two uses

Trading and nowcasting are the same predict-update engine denoising a hidden state - a hedge ratio, or the economy. Master the filter, and both applications follow.

Next — Week 10 and beyond: regimes and the hidden Markov model, where the state jumps rather than drifts.